NCI-Forex — NeuroCommand Intelligence Forex Trading Skill

Version: 3.0 | **Author:** SuperBrain Consolidated Logic

Classification: NCI Brain Logic Module — Live Trade Execution Capable

Target: MT4 / MT5 Brokers via NCI Trade Bridge

SECTION 0 — BRAIN IDENTITY & PURPOSE

This skill teaches the NCI NeuroCommand Brain how to trade the Forex market with institutional-grade confluence logic. It does not operate as a static Expert Advisor. It operates as **living brain logic** — reading market conditions, scoring opportunities, managing risk dynamically, and executing trades with full awareness of structure, momentum, volume, correlation, and multi-timeframe context.

The NCI brain should treat this document as its **trading nervous system**. Every section below is a layer of that nervous system.

Core Philosophy

- **Many entry doors. One risk manager.** The brain has 8 independent entry ports. Any port can open a trade. Once open, a single unified dynamic stop loss system owns the trade until it closes.
- **Confluence over confirmation.** No single signal is enough. The brain scores signals and only acts when enough align.
- **The stop loss is alive.** It moves, tightens, and anchors to real market structure — never static, never wide.
- **Correlation is context.** The brain always checks whether peer markets agree before committing.

SECTION 1 — MARKET READING LAYERS

The brain reads the market through 7 independent perception layers. Each layer answers one question.

Layer 1 — Trend Direction (Macro)

Question: What is the dominant trend direction?

Method — SMA Stack (200/50):

- `SMA(50) > SMA(200)` → Macro BULL
- `SMA(50) < SMA(200)` → Macro BEAR
- Equal or crossing → NEUTRAL / Caution

Method — KAMA Fan Alignment (10 EMA periods: 5,10,20,30,40,50,60,70,80,100):

- Count how many consecutive pairs are: `fast EMA > slow EMA`
- If ≥ 67% align ascending → BULL regime confirmed
- If ≥ 67% align descending → BEAR regime confirmed
- Below threshold → No regime (reduce position sizing)

Method — Trend Cloud (KAMA Base):

- KAMA(100) base line slope rising → Macro bull
- KAMA(100) base line slope falling → Macro bear
- Use 18 KAMA lengths (5 through 100); percentage in ascending order = trend strength score

Layer 2 — Market Structure (Swing Context)

Question: Is the market making higher highs or lower lows?

Method — Confirmed Pivot Structure (Lookback = 10 bars each side):

Pivot High detected when: bar[N] high is highest of bars[N-10] through bars[N+10]
Pivot Low detected when: bar[N] low is lowest of bars[N-10] through bars[N+10]

Structure States:

- New pivot high > last pivot high → `HH` → TREND_UP
- New pivot high < last pivot high → `LH` → TREND_DOWN
- New pivot low > last pivot low → `HL` → TREND_UP
- New pivot low < last pivot low → `LL` → TREND_DOWN

Rule: Structure only confirmed on the bar AFTER the pivot lookback window closes. Never trade live/repainting structure.

Layer 3 — Momentum & Energy

Question: Is there real force behind this move?

Method A — Kinetic Energy Score:

```
kinetic_energy[bar] = volume[bar] × |close[bar] - close[bar-1]|
```

Normalize over last 100 bars to 0.0–1.0 scale:

```
normalized = (kinetic_now - min_100) / (max_100 - min_100)
```

- Score ⩾ 0.45 → Significant momentum (entry confidence +1)
- Score ⩾ 0.75 → Major momentum (kinetic spike — record level for SL anchoring)

Method B — RSI Position:

- RSI(14) crossing above 30 → oversold bounce momentum
- RSI(14) crossing below 70 → overbought exhaustion momentum
- RSI(14) position relative to 50 → directional bias

Method C — Major Move Patterns:

```
avgBody = SMA(|close - open|, 50 bars)

LongBullBar = close > open AND (close - open) > avgBody × 1.5
LongBearBar = close < open AND (open - close) > avgBody × 1.5

ThreeBullBars = LongBullBar[0] AND LongBullBar[1] AND LongBullBar[2]
ThreeBearBars = LongBearBar[0] AND LongBearBar[1] AND LongBearBar[2]

BullishEngulf = close > open AND close[-1] < open[-1] AND close > open[-1] AND
open < close[-1]
BearishEngulf = close < open AND close[-1] > open[-1] AND close < open[-1] AND
open > close[-1]

MajorBullMove = ThreeBullBars OR BullishEngulf
MajorBearMove = ThreeBearBars OR BearishEngulf
```

Layer 4 — Support & Resistance Intelligence

Question: Where are the real walls — and are they strong?

Method — Logistic Regression Scored S/R Zones:

For every confirmed pivot (PivotLength = 14), score it with 2 binary features:

```
rsi_binary  = (RSI at pivot > 50) ? +1 : -1
body_binary = (|body| > ATR at pivot) ? +1 : -1
```

Run logistic regression model across all pivots of same type:

```
z = -(bias + w1 × rsi_binary + w2 × body_binary)
probability = 1 / (1 + exp(z))
```

Weights updated via gradient descent (lr = 0.008) as each zone is respected or broken.

Zone is ACTIVE if: `probability ≥ 0.70` AND zone has not been broken by a close.

Zone Events (trigger entry ports):

Event	Condition	Signal
Support Retest	`low < level` AND `close > level`	BUY bounce
Resistance Retest	`high > level` AND `close < level`	SELL bounce
Resistance Break	`high > level` AND `close ≥ level`	BUY breakout
Support Break	`low < level` AND `close ≤ level`	SELL breakdown

Zone Use in SL: Nearest active support below price = preferred BUY stop anchor. Nearest active resistance above price = preferred SELL stop anchor.

Layer 5 — Channel & Volatility

Question: Is price breaking out with volatility conviction?

Method — Smart Trend Flow Channel:

```
upper_channel = Highest(High, 20 bars)[confirmed]
lower_channel = Lowest(Low, 20 bars)[confirmed]
basis = (upper + lower) / 2
```

Bull breakout: `close > upper + ATR×0.1` AND `close[-1] ≤ upper[-1] + ATR×0.1`
Bear breakout: `close < lower - ATR×0.1` AND `close[-1] ≥ lower[-1] - ATR×0.1`

Volatility Gate — Z-Score:

```
chWidth = (upper - lower) / basis × 100
zScore = (chWidth - SMA(chWidth, 20)) / StdDev(chWidth, 20)
sigmoidVol = 100 / (1 + exp(-zScore))
```

- `sigmoidVol > 60` → High volatility (good for breakouts)
- `sigmoidVol < 30` → Low volatility (good for bounces, avoid breakouts)

Layer 6 — Higher Timeframe Confirmation

Question: Does the bigger picture agree?

Method — Non-Repainting HTF Check (×4 multiplier):

- Chart on H1 → check H4. Chart on M15 → check H1.
- Always read the CLOSED HTF bar (shift = 1), never the live bar.

HTF Signals:

```
HTF_EMA_Bull = EMA(20, HTF) > EMA(50, HTF)
HTF_Channel_Bull = HTF_close > HTF_Highest(High, 20) × 0.99

HTF_Confirm_Bull = HTF_EMA_Bull AND HTF_Channel_Bull
HTF_Confirm_Bear = NOT HTF_EMA_Bull AND HTF_close < HTF_Lowest(Low, 20) × 1.01
```

Layer 7 — Correlation Intelligence

Question: Are related markets moving in the same direction?

Method — Pearson Correlation (100-bar rolling):

```
r = Pearson(Symbol_close[], Peer_close[], 100)
```

Correlation Voting Rules:

- `|r| < 0.50` → Peer is unrelated, skip vote
- `r ≥ 0.50` → Positive correlation: peer BULL = agree with our BULL signal
- `r ≤ -0.50` → Negative correlation: peer BULL = agree with our BEAR signal (inverse)

Agreement Threshold: ⩾ 67% of related peers must agree for correlation bonus.

Default Peer Set (Forex): EURUSD, GBPUSD, AUDUSD
(NCI brain can expand this set based on the symbol being traded)

SECTION 2 — THE 8 ENTRY PORTS

The brain has 8 independent entry ports. Each port scores either +1 BUY or +1 SELL toward the total confluence score. The brain trades when the score reaches the minimum threshold.

Scoring Table

Port #	Name	Source	BUY condition	SELL condition
1	Channel Breakout	Smart Trend Flow	Close breaks above 20-bar high	Close breaks below 20-bar low
2	Kinetic Conviction	Kinetic Momentum Vectors	Normalized kinetic ⩾ 0.45 AND price > EMA(100)	Normalized kinetic ⩾ 0.45 AND price < EMA(100)
3	PRISM Dual SuperTrend	Trend Architect	Both ST rails flip bullish	Both ST rails flip bearish
4	KAMA Regime Gate	Trend Architect	⩾ 67% of 10 EMAs ascending	⩾ 67% of 10 EMAs descending
5	MA Ribbon	Trend Architect	EMA(20) > EMA(25)	EMA(20) < EMA(25)
6	HTF Confirmation	PineCoders HTF	HTF EMA bull AND HTF channel bull	HTF EMA bear AND HTF channel bear
7	S/R LR Zone	Flux Charts LR	Support retest OR resistance break	Resistance retest OR support break
8	Peer Correlation	TradingFinder	⩾ 67% of related peers confirm bull	⩾ 67% of related peers confirm bear

Additional Entry Port: AI Brain RSI + Structure + Major Move

(Bonus port — adds to score if conditions met)

```
BUY: RSI(14) crosses above 30
     AND SMA(50) > SMA(200)
     AND Structure == TREND_UP
     AND (ThreeBullBars OR BullishEngulf)  [optional: UseMajorMoves = true]

SELL: RSI(14) crosses below 70
      AND SMA(50) < SMA(200)
      AND Structure == TREND_DOWN
      AND (ThreeBearBars OR BearishEngulf)  [optional]
```

Minimum Entry Threshold

```
DEFAULT: MinSignals = 4 out of 8 (or 9 with AI Brain port)
CONSERVATIVE: MinSignals = 5
AGGRESSIVE: MinSignals = 3
```

Hard Guards (block trade even if score met)

The brain must reject an entry if any hard guard fires:

Guard	Description	Overridable?
ReqKinetic	Kinetic energy must be present	Yes (set false)
ReqRegime	KAMA regime must align	Yes (set false)
ReqHTF	HTF must confirm	No (default off unless enabled)
ReqSR	S/R zone interaction must be present	No (default off)
CooldownBars	Minimum bars between entries = 3	Always active
Conflict	Bull AND bear score equal → no trade	Always active

SECTION 3 — DUAL SUPERTREND PRISM RAIL LOGIC

The PRISM system uses TWO SuperTrend rails. Both must agree for a signal.

Rail 1 (Alpha — faster):

- ATR Period: 10

- Factor: 0.2

Rail 2 (Sigma — slower):

- ATR Period: 20
- Factor: 0.5

SuperTrend Band Carry Rule (critical for accuracy):

```
upper_band[i] = IF rawUpper < upper_band[i+1] OR close[i+1] > upper_band[i+1]
                THEN rawUpper
                ELSE upper_band[i+1]

lower_band[i] = IF rawLower > lower_band[i+1] OR close[i+1] < lower_band[i+1]
                THEN rawLower
                ELSE lower_band[i+1]
```

Direction flip:

```
IF was BEARISH: flip to BULLISH when close > upper_band
IF was BULLISH: flip to BEARISH when close < lower_band
```

PRISM Signal:

```
NOW_BULL = dir1 == BULLISH AND dir2 == BULLISH
NOW_BEAR = dir1 == BEARISH AND dir2 == BEARISH
WAS_BULL = dir1[-1] == BULLISH AND dir2[-1] == BULLISH
WAS_BEAR = dir1[-1] == BEARISH AND dir2[-1] == BEARISH

BullFlip = NOW_BULL AND NOT WAS_BULL
BearFlip = NOW_BEAR AND NOT WAS_BEAR
```

Confidence Scoring for PRISM Flip (simplified LR):

```
Features:
  rsiF    = (RSI(14) - 50) / 50          → range [-1, +1]
  bbF     = (close - BB_mid) / (2 × BB_dev)  → range [-1, +1]
  momF    = sum(consecutive candle direction, 10) / 10
  atrF    = (ATR_now / ATR_20bars_ago) - 1.0
  stDistF = (close - ST_line) / (ATR × 3)

Fixed weights: RSI=0.80, BB=0.60, Momentum=0.50, STDist=0.40, ATRSlope=0.30

z = sign × (0.80×rsiF + 0.60×bbF + 0.50×momF + 0.40×stDistF) - 0.30×atrF
confidence = sigmoid(z) × 100%
```

Flip only counts toward entry score if `confidence ≥ MinConfidence` (default 55%).

SECTION 4 — VOLUME PROFILE INTELLIGENCE

The brain builds a mental volume profile over the last 400 bars, divided into 30 bins.

For each price bin:

```
bin_volume[i] += volume[bar] for all bars where close falls in bin[i]
buy_volume[i] += volume × buyPercent  [buyPercent = (close - low)/(high - low)]
```

High Volume Node (HVN) Detection:

```
HVN = bin[i] where bin_volume[i] > bin_volume[i-1] AND > bin_volume[i+1]
    → local volume peak = strong support/resistance
```

Volume Gap (Void) Detection:

```
Gap = bin[i] where (maxVol - bin_volume[i]) / maxVol > 0.60
    → less than 40% of max volume = price moves fast through here
```

Usage in Entry Logic:

- Price approaching HVN from below → BUY anticipation zone
- Price approaching HVN from above → SELL anticipation zone
- Price in Volume Gap → reduced friction, trend continuation likely

- Alert when price enters within `HVN_ProxPct × range` of HVN level

Usage in SL Logic:

- HVN below entry = natural support → prefer SL just below HVN
- HVN above entry = natural resistance → prefer SELL SL just above HVN

SECTION 5 — LIQUIDITY LEVELS (PIVOT-BASED)

The brain tracks up to 5 pivot highs and 5 pivot lows as liquidity reference points.

Detection (PivLeft = 20, PivRight = 20):

```
Pivot High: bar[N] high is highest of all bars in [N-20, N+20]
Pivot Low:  bar[N] low  is lowest  of all bars in [N-20, N+20]
```

Usage:

- Nearest pivot high above price = liquidity resistance → SELL target or SELL SL anchor
- Nearest pivot low below price = liquidity support → BUY target or BUY SL anchor
- Break above pivot high with volume → BUY entry trigger (resistance cleared)
- Break below pivot low with volume → SELL entry trigger (support cleared)

SECTION 6 — SESSION AWARENESS

The brain understands which trading session is active and adjusts behavior accordingly.

Session	UTC Time	Behavior
Tokyo	23:00–07:00	Lower volatility. Favor bounce/retest entries. Reduce lot size 20%.
London	07:00–16:00 (BST: 06:00–15:00)	High volatility. All entry types valid. Full position size.
New York	12:00–20:00 (EDT: 13:00–21:00)	High volatility. Trend continuation strongest. Full position size.
London/NY Overlap	12:00–16:00	Peak liquidity. Breakout entries strongest.
Off-hours	Outside all sessions	No new entries. Manage existing trades only.

DST Handling: London DST shifts all times 1 hour earlier. New York DST shifts 1 hour earlier.

SECTION 7 — THE DYNAMIC STOP LOSS ENGINE

This is the most critical part of the skill. The brain manages one stop loss per open trade. It only moves in the direction of profit. It never widens.

Stop Loss Sources (priority order for initial SL)

1. **LR S/R Zone** — If an active LR-scored support zone exists below entry (for BUY), use `zone_level - ATR×0.3`
2. **Kinetic Spike Level** — Last confirmed kinetic spike support level
3. **Pivot Low** — Nearest confirmed pivot low below entry
4. **ATR Initial** — `entry - ATR(14)×2.0` as the fallback

Stop Loss Trail Methods (Mode 4 = All Combined)

Method 1 — Channel Band Trail:

```
For BUY: SL_candidate = Lowest(Low, 20) - ATR×0.3
For SELL: SL_candidate = Highest(High, 20) + ATR×0.3
```

Method 2 — Kinetic Spike Level:

```
For BUY:  SL_candidate = last_kinetic_support - ATR×0.3
For SELL: SL_candidate = last_kinetic_resistance + ATR×0.3
```

Method 3 — ATR Step Trail:

```
For BUY:  SL_candidate = Bid - ATR×0.5
For SELL: SL_candidate = Ask + ATR×0.5
```

Method 4 — LR S/R Zone Anchor:

```
For BUY:  SL_candidate = nearest_active_support - ATR×0.3
For SELL: SL_candidate = nearest_active_resistance + ATR×0.3
```

Final SL Rule:

```
new_SL = MAX(all_SL_candidates)  [for BUY — never below current SL]
new_SL = MIN(all_SL_candidates)  [for SELL — never above current SL]

NEVER widen:
  BUY: new_SL < current_SL → reject
  SELL: new_SL > current_SL → reject
```

Breakeven Promotion:

```
When TP1 is hit → move SL to entry + ATR×0.3 (breakeven buffer)
```

Profit Target Tiers (Scale-Out Strategy)

```
TP1 = entry ± ATR×1.5  → Close 40% of position
TP2 = entry ± ATR×2.5  → Close 40% of remaining
TP3 = entry ± ATR×4.0  → Let remainder run to full target (broker TP order)
```

SECTION 8 — PATTERN DETECTION AWARENESS

The brain recognizes classical chart patterns and uses them to strengthen entry conviction. Patterns are never the primary entry signal — they are context enhancers.

Recognized Patterns and Their Bias

Pattern	Type	Bias	Target Method
Double Top	Reversal	BEAR	Height of pattern projected downward from neckline
Double Bottom	Reversal	BULL	Height of pattern projected upward from neckline
Triple Top	Reversal	BEAR	Same as double top
Triple Bottom	Reversal	BULL	Same as double bottom
Head & Shoulders	Reversal	BEAR	Head height from neckline projected down
Inv. H&S	Reversal	BULL	Head depth from neckline projected up
Bullish Flag	Continuation	BULL	Pole height added to breakout point
Bearish Flag	Continuation	BEAR	Pole height subtracted from breakout point
Bullish Pennant	Continuation	BULL	Pole height from apex
Bearish Pennant	Continuation	BEAR	Pole height from apex
Rising Wedge	Reversal	BEAR	Base width from breakout
Falling Wedge	Reversal	BULL	Base width from breakout
Ascending Triangle	Continuation	BULL	Base height from breakout
Descending Triangle	Continuation	BEAR	Base height from breakout
Cup & Handle	Continuation	BULL	Cup depth added to rim breakout
Inv. Cup & Handle	Continuation	BEAR	Cup depth subtracted from rim breakdown

Pattern Validation Rules:

```
min_size = MAX(price × 0.005, ATR(14) × 1.0)   [pattern must be meaningful]
sym_tol  = 0.10  [shoulder symmetry tolerance = 10%]
lvl_tol  = 0.03  [level equality tolerance = 3%]

Breakout confirmed when: close beyond level by ATR×1.0 (or % threshold)
Volume confirmation: breakout bar volume > SMA(volume, 20) × 1.0 (optional)
Min R:R: 1.0 (default) — pattern target must be worth the risk
```

SECTION 9 — RISK MANAGEMENT PARAMETERS

Position Sizing

```
Risk per trade: 1.0% of account balance (adjustable)
Lot size = (balance × risk%) / (SL_pips × pip_value)
Max lots: 5.0 (hard cap)
Lot step: rounded to broker's minimum step
```

Drawdown Protection

```
If account drawdown > 10% from peak: reduce RiskPct to 0.5%
If account drawdown > 15% from peak: pause new entries, manage only
If account drawdown > 20% from peak: close all, alert, wait for human review
```

Spread Filter

```
Do not enter if spread > ATR(14) × 0.2
Do not enter if spread > symbol's average spread × 3.0
```

Slippage Tolerance

```
Max acceptable slippage: 3 pips (broker order instruction)
```

SECTION 10 — NCI BRAIN DECISION LOGIC (PSEUDOCODE)

This is the complete decision loop the NCI brain runs on every closed bar.

```
ON_NEW_BAR_CLOSED:

  // STEP 1: Update all perception layers
  UPDATE Structure(lookback=10)
  UPDATE KAMA_Fan(periods=[5,10,20,30,40,50,60,70,80,100])
  UPDATE Kinetic_Energy(norm_lookback=100)
  UPDATE SR_Zones(pivot_len=14, lr_threshold=0.70)
  UPDATE Channel(length=20)
  UPDATE PRISM(ST1_period=10, ST1_factor=0.2, ST2_period=20, ST2_factor=0.5)
  UPDATE HTF(mult=4, always_use_closed_bar=TRUE)
  UPDATE Correlation(peers=[EURUSD,GBPUSD,AUDUSD], period=100)
  UPDATE Volume_Profile(lookback=400, bins=30)
  UPDATE Session_State()

  // STEP 2: If trade is already open → manage only
  IF has_open_trade:
    CHECK TP1, TP2 partial close conditions
    CALCULATE new_SL from all 4 trail methods
    IF new_SL improves: MODIFY stop loss
    RETURN (do not look for new entry)

  // STEP 3: Check cooldown
  IF bars_since_last_signal < 3: RETURN

  // STEP 4: Score signals
  bull_score = 0
  bear_score = 0

  IF channel_bull_break:  bull_score++
  IF channel_bear_break:  bear_score++

  IF kinetic_bull:        bull_score++
  IF kinetic_bear:        bear_score++

  IF prism_bull AND confidence >= 55%: bull_score++
  IF prism_bear AND confidence >= 55%: bear_score++

  IF kama_regime_bull:    bull_score++
  IF kama_regime_bear:    bear_score++

  IF ribbon_bull:         bull_score++
  IF ribbon_bear:         bear_score++
```

```
IF htf_confirm_bull:    bull_score++
IF htf_confirm_bear:    bear_score++

IF sr_zone_event == BUY:   bull_score++
IF sr_zone_event == SELL:  bear_score++

IF correlation_agrees_bull: bull_score++
IF correlation_agrees_bear: bear_score++

// AI Brain bonus port
IF ai_brain_signal == BUY:  bull_score++
IF ai_brain_signal == SELL: bear_score++

// STEP 5: Apply hard guards
IF ReqKinetic AND NOT kinetic_present:
  bull_score = MIN(bull_score, MinSignals-1)
  bear_score = MIN(bear_score, MinSignals-1)

IF ReqRegime AND NOT regime_aligned:
  [same reduction]

// STEP 6: Entry decision
IF bull_score >= MinSignals AND bull_score > bear_score:
  OPEN BUY with:
    SL = best_of(LR_zone, kinetic_support, pivot_low, ATR×2.0)
    TP1 = entry + ATR×1.5
    TP2 = entry + ATR×2.5
    TP3 = entry + ATR×4.0
    lots = CalcLots(SL_distance, RiskPct=1.0%)
    LOG reason string with all 8 port scores

ELIF bear_score >= MinSignals AND bear_score > bull_score:
  OPEN SELL with:
    SL = best_of(LR_zone, kinetic_resistance, pivot_high, ATR×2.0)
    [mirror of BUY logic]
    LOG reason string

ELSE:
  NO TRADE this bar
```

SECTION 11 — ENTRY PORT REFERENCE CARD

Quick reference for the NCI brain to call each port.

```
PORT 1:  EntrySignal_ChannelBreakout(symbol, tf, len=20, buf_atr=0.1)
PORT 2:  EntrySignal_KineticMomentum(symbol, tf, norm=100, thresh=0.45, ema=100)
PORT 3:  EntrySignal_PRISM_DualST(symbol, tf, f1=0.2, p1=10, f2=0.5, p2=20)
PORT 4:  EntrySignal_KAMA_Regime(symbol, tf, align_pct=0.67)
PORT 5:  EntrySignal_MARibbon(symbol, tf, len=20)
PORT 6:  EntrySignal_HTF_Confirm(symbol, tf, mult=4)
PORT 7:  EntrySignal_SRLR_LogisticSR(symbol, tf, piv=14, prob=0.70)
PORT 8:  EntrySignal_Correlation(symbol, tf, peers[], period=100, min_r=0.50)
PORT 9:  EntrySignal_AIBrain_RSI_Structure_MajorMoves(symbol, tf)

SL ENGINE: DynamicSL_Engine(mode=4, trail_atr=0.5, buffer_atr=0.3)
```

SECTION 12 — INDICATOR AWARENESS CATALOGUE

All indicators the NCI brain is aware of and can compute natively.

Indicator	Type	Use
RSI(14)	Momentum	Overbought/oversold, cross triggers, LR feature
SMA(50), SMA(200)	Trend	Macro trend filter
ATR(14)	Volatility	SL sizing, band calculations, kinetic scaling
EMA(20), EMA(25), EMA(50), EMA(100)	Trend	Ribbon, regime, baseline
KAMA(5–100, 10 periods)	Adaptive MA	Regime fan, trend cloud
SuperTrend(0.2/10, 0.5/20)	Trend	PRISM dual rail
Pivot High/Low (lookback 10–20)	Structure	Structure trend, LR zones, SL anchor
Highest(High, 20) / Lowest(Low, 20)	Channel	Breakout detection, HTF channel
Volume SMA(20)	Volume	Kinetic energy reference, spike filter
Pearson Correlation (100-bar)	Statistical	Peer agreement scoring
Logistic Regression (2-feature)	ML	S/R zone quality scoring
Polynomial Regression (degree 4)	ML	PRISM lookback optimization
Bollinger Bands(20, 2σ)	Volatility	LR feature, BB position signal
Hurst Exponent (100-bar)	Statistical	Regime trending/mean-reverting detection
ALMA(20, 0.85, 6.0)	Adaptive MA	Ribbon approximation
Stochastic RSI(14, 14, 3)	Momentum	Super Channel composite
MFI(14)	Volume-Momentum	Super Channel composite
CCI(20) / Percentrank	Momentum	Super Channel composite
Keltner Channel(20, 2.0)	Volatility	Super Channel composite

SECTION 13 — WHAT THE NCI BRAIN MUST NEVER DO

1. **Never widen a stop loss.** Moving SL away from price is forbidden once set.
2. **Never enter during zero-liquidity periods** (broker closed, news blackout, spread > ATR×0.2).
3. **Never trade when bull_score == bear_score** (conflicting signals = no trade).
4. **Never use live/unconfirmed bar data** for structure or pivot decisions. Always bar[1] minimum.
5. **Never enter a second trade while one is already open** (one position at a time, unified management).
6. **Never ignore the correlation filter** when 2+ related peers actively disagree.
7. **Never skip the cooldown period** (minimum 3 closed bars between entries).
8. **Never over-ride the drawdown protection** rules without explicit human instruction.

SECTION 14 — SKILL CHANGELOG

Version	Change
1.0	Initial: Channel + Kinetic + PRISM + KAMA + Ribbon (5 ports)
2.0	Added: HTF Confirmation + LR S/R Zones + Correlation Filter (8 ports)
3.0	Added: AI Brain RSI+Structure+MajorMove (9th port), Volume Profile HVN/Gap, Liquidity Pivots, Session Awareness, Pattern Context, Full Pseudocode Decision Loop

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NCI-Forex.md — End of Document

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